

MUHAMMAD ISLAM

Testing Engineer – Protection & Control

Address: 4th St, 8741, Al-Thuqbah, Al Khobar, 34623, Saudi Arabia

PROFILE SUMMARY

Electrical Engineer specializing in **Protection & Control**, Testing & Commissioning, and substation systems across transmission, distribution, and power generation networks. Experienced in relay testing, configuration, troubleshooting, and panel functional testing. Strong analytical skills with the ability to work under pressure in energized environments. Known for delivering accurate technical reports, maintaining high safety standards, and supporting successful project execution.

Mabran Al-Arabia Contracting Company -KSA (2024 – till dated)

Joined Mabran AL Arabia as a Field Services Engineer and responsible for installation, testing & commissioning activities.

ABB Project Execution (1.5 Years)

Performed project activities using ABB's project platform, focusing on documentation, reporting, and workflow coordination.

Protection Relay Testing – Multi-Plant Scope

Plants:

RTR – ASHOMCo, Juaymah ARAMCO, MARAFIQ GT-23/31/33, Shedgum Gas Plant – ARAMCO

Scope of Work:

- G60A – Generator Primary Protection Testing
- G60B – Generator Backup Protection Testing
- T60 – Unit Transformer Protection
- T60T – Main Transformer Protection
- C60 – Breaker Failure Relay
- Nexus GEN DMM Meter testing
- Transducer testing for generator measurement

New Substation – Scheme Check, Functional Testing & Relay/Switchgear Testing

34.5 kV & 13.8 kV GIS | Gulf Cryo – Ras Al Khair

Key Responsibilities

- Performed **complete scheme checks** for the new substation, including protection logic, breaker control circuits, interlocks, and trip/close sequences
- Conducted **functional testing** for all protection and control circuits across 34.5 kV and 13.8 kV GIS bays
- Executed **relay testing** for feeder, transformer and busbar ensuring correct pickup, timing, and trip logic
- Carried out **switchgear testing** including breaker timing, contact resistance, insulation resistance, and mechanical/electrical interlock verification
- Verified **CT/VT circuits**, polarity, ratio, wiring integrity, and relay input correctness for all GIS bays.
- Conducted transformer stability testing for differential protection accuracy

Protection Relay Testing & CT Replacement – 13.8 kV & 34.5 kV GIS

Khafji Joint Operations (KJO) | Oct 12, 2025 – July 15, 2026

34.5 kV Feeder Protection Relay Upgrade

Key Responsibilities:

- Tested and commissioned SEL-311, SEL-411L, SEL-787, SEL-735, and SEL-451 protection relays
- Replaced CTs in 34.5 kV GIS bays with full testing and commissioning
- Conducted secondary injection, line stability, transformer stability, and bus bar stability tests
- Verified breaker control circuits and relay functionality

13.8 kV Feeder Protection Relay Upgrade

• Key Responsibilities:

- Replaced old feeder protection relays and modified associated control wiring.
- Installed, configured, and commissioned **SEL-751** Feeder Protection Relays for **Bus A & B**.
- Performed secondary injection testing, and breaker close/trip/interlock verification.
- Verified earth switch and CB interlocking operations and status indications.
- Completed **Bus Bar** Stability checks during protection upgrade.
- Conducted full signal verification with **PSA** (Power System Automation)

Panel Commissioning – SABIC Al-Razi Plant 1 & 2 (15 Aug 2025)

Key Responsibilities:

- Commissioned ABB protection and control panels for generator incomer and two outgoing feeders.
- Performed relay testing, CT/VT scheme checks, and functional logic verification.
- Executed hot and cold commissioning of protection, control, and interlocking schemes.
- Completed end-to-end testing, wiring verification, and system readiness checks.
- Conducted busbar contact resistance, Hi-Pot insulation testing, and differential stability testing.

33kV Switchgear Pilot Differential protection upgrade at SEC SS7075,7328,7071,7031,7027 and 7054 in Riyadh KSA

Key Responsibilities:

- Dismantling of existing pilot differential relay and P142 control circuit modification,
- Installation of Protection panel and commissioned with existing breaker.
- CT, VT, testing and Secondary injection and primary Injection test.
- Scada open loop and Close loop and MFM Meter test.
- P543 Differential Protection relay and P142 OC/EF relay test.
- End to End test and final trip test.
- Function Control & Interlock Circuit, CB Closing & Tripping test.
- Earth switch & CB interlocking and Operation, status checked.

Testing Engineer – Protection & Control

EPESOL Pvt. Ltd. 2022 – 2024

- Tested generator, transformer, distance, and breaker-failure protection relays at 1160MW, 500kV QATPL HEI Power Plant.
- Performed switchyard relay testing at 220kV, 596MW UCH-I Power Plant.
- Executed setting modifications and testing of Schneider feeder relays (P111, P343, P341, P127, VAMP40) at Al-Moiz Sugar Mill.
- Conducted testing of distance, generator, feeder relays, PQM meters, and transducers at 730MW, 220kV Lalpir/Pakgen Power Plant.
- Performed functional testing of Generator Protection Panels (GPP) at EPESOL panel shop.
- Configured and tested feeder protection relays at Lalpir Power Plant.

Panel Testing at Factory (FAT)

- Performed FAT for 10 protection panels for JSPDI.
- Conducted FAT for 17 protection panels for QESCO.
- Tested 2 Generator Protection Panels (GPP) for Al-Moiz Sugar Mill.
- Tested 2 MP4 Metering Panels for MEPCO.

CERTIFICATIONS

- OHS General Induction
- Basic Fire Fighting
- Distribution Safety Rules & Safe Isolation / Switching Procedure
- National Grid Safe Working Procedure
- SEC ID
- Competent Work ID
- Confined Space Training
- Maaden Contractor – All Trainings Completed

PROTECTION RELAYS HANDLED

Line Protection

- SEL 311, SEL 411
- P543 (GE), D60 (GE)
- 7SA522, 7SD (Siemens)
- RED670 (ABB)
- Micromho (Areva/Alstom)
- LZ96A

Generator Protection

- G60, 489 (GE)
- P343, P345 (Alstom)
- REG 615 (ABB)

Transformer Protection

- SEL 787
- RET615, RET630 (ABB)
- T60 (GE)

Motor Protection

- 269, 369 (GE)
- REM615, REF542Plus (ABB)
- SEL 710

Feeder / Bay Protection

- SEL 751
- RED 615, REU620, REB615
- SP14NB
- MIF II (GE)
- 7SJ51X (Siemens)
- REF615 (ABB)
- P142 (Areva/Alstom)
- P122, P127 (Schneider)

TEST EQUIPMENT HANDLED

Secondary Injection Kits

- CMC256, CMC356 (Omicron)
- FREJA 300 / 306 (Programma / Megger)
- Megger Secondary Injection Set
- Severker 760

Primary Injection Kits

- CPC100 (Omicron)
- ISA 3000

Circuit Breaker Analyzer

- EGIL
- Megger Programma TM1600

Micro Ohm Meter

- MOM200

- Omicron Equipment
- CPC100
- CT Analyzer

Doble

- F6150 Protection Relay Test Set

SOFTWARES HANDLED

- ABB PCM600
- SEL Accelerator QuickSet
- Schneider Electric Easergy Studio
- Alstom MiCOM S1 Agile
- Siemens DIGSI
- GE Multilin / Enervista UR Setup
- OMICRON Test Universe
- Freja Win
- Doble F6150 Software